
Intelligence Engineering

Solving Real World Problems At Scale

Prof. Dr.-Ing. Mark Schutera | September 25, 2026 · *drifting stream*

LECTURE IV

Training Data

This lecture in one slide

IV Training Data

Sampling

Labeling

Class Imbalance

Data Augmentation

*why this lecture matters, and what
you can do after it*

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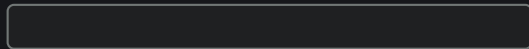
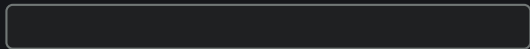
Sampling

Nonprobability Sampling

Definition: Nonprobability sampling

Simple Random Sampling

$$P(y | x)$$



Stratified Sampling

$$P(y|x)$$

$$[1]$$

Weighted Sampling

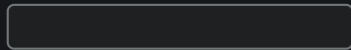
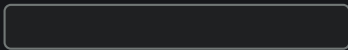
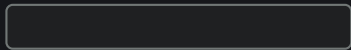
$P(y|x)$

Weighted sampling

[1]

Reservoir Sampling

$$P(y|x)$$



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Importance Sampling

$$P(y | x)$$

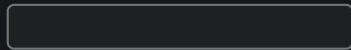
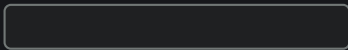
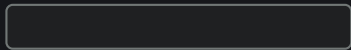
$$[1]$$

2

Labeling

Hand Labels

$P(\hat{y} | x)$



[1]

Label multiplicity

$$P(\hat{y} | x)$$

$$[1]$$

Data lineage

$P(y|x)$

Definition: Data lineage

[1]

[1]

Feedback loop length

$P(y|x)$

Short feedback loops

Long feedback loops

[1]

Handling the Lack of Labels

$P(y|x)$

Weak supervision

Semi-supervision

Transfer learning

Active learning

[1]

Weak supervision

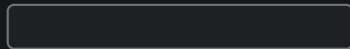
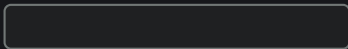
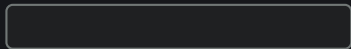
Weak supervision

$P(y | x)$

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Semi-supervision

$P(y|x)$



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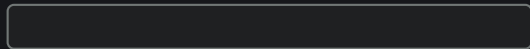
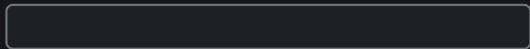
Transfer learning

$$P(\hat{y} | x)$$

[1]

Active learning

$$P(\hat{y} | x)$$



[1]

3

Class Imbalance

Challenges of Class Imbalance

$P(\hat{y} | x)$

Training Data

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Handling Class Imbalance

$P(y|x)$

Evaluation metrics

Data-level methods

**Algorithm-level
methods**

[1]

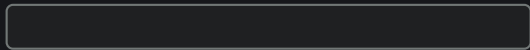
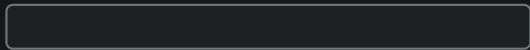
Using the right evaluation metrics

$P(\hat{y} | x)$

[1]

Data-level methods: Resampling

$P(y|x)$



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Algorithm-level methods

$P(\hat{y} | x)$

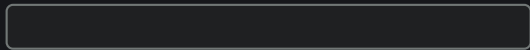
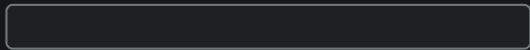
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Data Augmentation

Simple Label-Preserving Transformations

$P(y|x)$



Perturbation

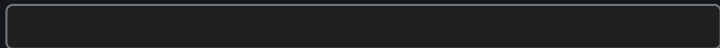
Perturbation

$P(y | x)$

[1]

Data Synthesis

$$P(y|x)$$



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Summary

Summary

$P(\hat{y} | x)$

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[1]

References

- 1 Chip Huyen. *Designing Machine Learning Systems: An Iterative Process for Production-Ready Applications*. Sebastopol, CA: O'Reilly Media, 2022.